Centralization Paradox of DAO: Comparative Study on AI Agent Protocol Standardization

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Abstract

Decentralized autonomous organizations (DAOs) propose a structurally distinct alternative to corporate hierarchy, yet empirical comparisons of the two on the same technical domain remain scarce. We exploit a theoretically matched pair from 2025: ERC-8004 (Trustless Agents), an EIP advanced by permissionless rough consensus, and Google A2A, an agent interoperability protocol now governed by an eight-seat corporate committee. We ask whether permissionless DAO governance produces a higher degree of decentralization than corporate hierarchy in practice. We collect 4,323 governance participation records, annotate each with an LLM-assisted pipeline, and analyse the data through joint topic discovery (BERTopic and Thematic-LM) and three layered networks (co-participation, discourse, socio-semantic). Findings shows that governance form shapes the thematic focus of deliberation, while both networks exhibit comparably high participation inequality and fragmented community structure. We discuss the centralization paradox of permissionless DAOs and the confounding role of open-source norms, and argue that decentralization should be treated as a design objective rather than a structural fact.

Keywords

DAO Governance, technology standardization, open-source governance, consortium governance, blockchain, AI protocol

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1 Introduction

The governance of shared technical standards determines who controls the trajectory of technology and on that terms [51]. Decentralized autonomous organizations (DAOs) proposed a structually distinct answer to this question: coordination through open participation, transparent discussion and rough consensus rather than appointed comittees and invisible communication [14, 49]. Proponents argue that this architecture distributes power more broadly and introduces permissionless entry; while critics observe that concentration in token ownership and participation anathy often produce elite control in practice [47, 52, 53]. Despite that theoretical attention has been predominantly piad to DAOs as a governance model, empirical evidence directly comparing DAOs to corporate governance on the same analitical measure remains absent from the literation.

The governance of AI protocal infrastructure is heavily code-based, providing an opportunity to fill in the gap. In 2025, two agent interoperability protocals emerged under contrasting governance regimes. ERC-8004, which belongs to Ethereum Improvement Proposals (EIP), defines a smart-contract interface standard for on-chain AI agent identity. It "proposes to use blockchains to discover, choose, and interact with agents across organizational boundaries without pre-existing trust" [17]. Its lifecycle follows the staged defind in EIP-1, and is advanced by rough consensus among forum participants and EIP editors, with no formal votes and permissionless implementations [16]. Google A2A introduces "An open protocol enabling communication and interoperability between opaque agentic applications" [2]. It was initiated by Google and donated to the Linux Foundation in June 2025, now governed by an eight-seat Technical Steering Committee (TSC) composed entirely of corporate representatives [3, 21]. Both protocals address the same technical domain of trustless AI agent interoperability, but demonstrates opposite ends of the governance spectrum.

Taking ERC-8004 and Google A2A as a theoretically matched pair, we ask:

RQ: Compared to corporate hierarchy, does the governance structure of a permissionless DAO really achieve a higher degree of decentralization?

We decompose this into three more specific sub-questions:

- **RQ1 (Decision Architecture):** How do the formal decision procedures, entry rights, and authority structures of the two regimes differ?
- **RQ2 (Discourse Composition):** How does governance form shape the topical and argumentative composition of participation discourse?
- **RQ3 (Relational Networks):** How does governance shape co-participation ties, discourse-level consensus/conflict structures, and actor–topic divisions of collaborative labor?

To answer these questions, we first reconstruct the two governance mechanisms from their official documents and visualize them as decision-flow diagrams. We then collect 4,323 governance participation records from their public repositories and apply a Large Language Model (LLM)-assisted annotation pipeline to label each record's argumentative function, stance, and stakeholder affiliation [10]. On top of this annotation we run two topic-discovery

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methods: unsupervised BERTopic [22], and LLM-inductive Thematic-LM [50]. Moreover, we run three network analyses: co-participation social network analysis (SNA) [5], discourse network analysis [33], and socio-semantic bipartite networks [48]. The methodological triad mirrors the three sub-questions.

Three findings emerge. First, the two cases instantiate structurally opposite decision architectures: ERC-8004 advances by rough consensus with permissionless deployment, while A2A vests binding authority in a corporate technical steering committee. Second, governance form shapes the thematic focus of deliberation rather than its inequality: DAO governance concentrates discourse on security mechanisms and protocol principles, whereas corporate governance distributes deliberation across engineering-execution workstreams. Third, both networks exhibit comparably high participation inequality and fragmented community structure; discourse congruence is nonetheless denser in ERC-8004, reflecting tighter within-community consensus formation.

This paper makes three contributions. Empirically, it provides the first matched-case, multi-method comparison of DAO and corporate governance of an AI technology standard using actual participation data. Methodologically, it integrates LLM-assisted coding with topic-discovery and network-discourse methods in a single comparative pipeline, extending CSCW research on online deliberation in standardization processes [27], comparative discourse analysis of online communities [54], governance dynamics on open-source platforms [20, 34], and the combined institutional–socio-technical analysis of open-source collaboration [57]. Theoretically, it carries the debate on whether decentralization is a "revolution or a mirage" [47] into AI protocol standardization, where governance choices shape the trajectory of socio-technical infrastructure [55].

2 Related Work

Three streams of literature converge on this study.

Decentralized governance via blockchain. First, Harvey provides a comprehensive overview of the technologies, theories, and practices inherent to decentralized finance (DeFi) and blockchain technology [24]. Beck et al. [6] provided the foundational Information System framework mapping blockchain governance along decision rights, accountability and incentives dimensions. Ziolkowski et al. [59] identified six major governance challenges specific to the blockchain system, and half of them are unique to this technology. Kiayias and Lazos [28] systematised the field with a SoK review of blockchain governance mechanisms. More recently, Ellinger et al. [14] explored the balanced polycentric governance of digital commons. Reineke et al. [47] devloped an integrative theoretical framework, showing the evolution of decentralization concepts. Sunyaev et al. [55] called for purposeful, design-oriented decentralisation rather then ending at ideology. Motea and Oba [40] iterrogated the democratic legitimacy of blockchain governance structures.

Governance of decentralized versus corporate structure. Murray et al. [41] examined how smart contracts and DAOs alter agency costs in corporate contracting, providing leading opinions. Lumineau et al. [36] highlights tacitness of transactions and social implications of this technology. Lee [32] showed that decentralization is an onging process rather than a one-time change. Rahman et al. [45] and Hunt et al. [26] analyzed the dynamics and power accumulation of platforms. Hui and Tucker [25] address AI along with decentralized ecosystem, proposing an innovative governance framework. However, there studies rely on theoretical or interview-based methods; none of them use governance-participation data or computational methods to test structual difference between DAO and corporate governance in the same domain.

Computational studies of cooperative work. Researchers has long examined coordination in open, online communities. Mockus et al. [39] documented participation inequality in Apache and Mozilla; Crowston and Howison [13] showed hierarchy and centralization in free and open-source software team communications. Within the CSCW tradition specifically, Im et al. [27] analysed Wikipedia's Requests for Comments (RfC), a deliberation mechanism structurally analogous to EIP rough consensus, and revealed persistent imbalance between deliberation and resolution. Germonprez et al. [20] catalogued structural realities of contemporary open-source projects, including pervasive corporate engagement; Yin et al. [57] combined institutional analysis with socio-technical network methods to study open-source sustainability, providing a methodological precedent for our dual-lens design; Li et al. [34] examined code-of-conduct conversations on GitHub as a window into informal governance norms in open-source repositories. Kulakowski and Frasincar [30] introduced CryptoBERT, a domain-adapted BERT variant pre-trained on 3.2 million cryptocurrency social-media posts, establishing a specialised embedding backbone for blockchain-native corpora. More recently, Wu et al. [44] applied sentiment and discourse analysis to six DAO forums; Stine and Agarwal [54] proposed comparative discourse analysis via topic models; Shi et al. [50] introduced Thematic-LM for LLM-assisted inductive thematic analysis of large corpora; Ao et al. [5] used social network analysis on on-chain Aave data to show voting-power concentration; Leifeld [33] proposed discourse network analysis concentrating on participants' stance; Roth and Cointet [48] connets sematic analysis with social topology. Wang el al. [56] found the threat of power concentration via scaled empirical analysis of scale *SnapShot* data. Özdemir Sönmez et al. [53] quantified voting power concentration and participation apathy across DAO governance models, especially in token-based organizations. Chen et al. [11] applied a quasi-experimental PSM-DID design to 98,000 Steemit users, finding that governance token ownership boosts curation effort but reduces creation novelty. Carlson and Burbano [10] established foundational guidelines for LLM-assisted annotation in management research.

Table 1 positions sixteen representative works into three classes by paper type, each evaluated by criteria appropriate to that class. Panel (a) lists *foundational works* that originated the methods or concepts we inherit; panel (b) lists *perspective papers* that advance theoretical claims about DAO and corporate governance without conducting their own empirical tests; panel (c) lists *empirical and methodological studies* of online cooperative work, evaluated additionally by openness and whether they perform a cross-governance comparison. Panel (b) lacks empirical tests of its claims, while no panel-(c) study performs a matched-case comparison of DAO and corporate governance on the same technical domain—the joint gap this study closes by contributing 4,323 governance records across

ERC-8004 (permissionless DAO) and Google A2A (corporate TSC), with open data and analysis code.

3 Methodology

3.1 Comparative Case Study Design

We adopt a comparative case study design to examine how governance structure shapes participation dynamics in AI protocal standarization [58]. Interoperability is an inherited characteristic of decentralized applications (dApps) in DeFi [23], and ERC-8004 as an protocal specifically degisned for AI agents extends this feature. Google A2A, on the corporation's side, realized similar functions in different contexts. Moreover, they are originated within the same year (2025). Nevertheless, they differ sharply in governance architexture. ERC-8004 belongs to Ethereum, a permissionless and community-driven open-source DAO, and Google A2A is now governed by TSC. Therefore, the two protocals constitute a theoretically matched pair. Holding the technical domain constant while varying the governance form enables structured comparison of participation patterns, discourse composition, and network analysis.

3.2 Data Collection and LLM Annotation

3.2.1 Data sources. We collect governance participation records from the public repositories. For ERC-8004, data originate from two sourcess: (1) 113 posts from the Ethereum Magician forum thread [17]; and (2) 36 GitHub records from the nine pull requests that directly modified `ERCS/erc-8004.md` [15]. For Google A2A, data originate from three streams in the `a2aproject/A2A` repository [2]: (1) 3,104 issue and issue-comment records, (2) 1,955 pull requests and review-comment records, and (3) 822 GitHub Discussion records.

The raw records totalled 6,030. We removed the following data: records with fewer than 20 characters of body text, which primarily are CI notifications, merge-conflict markers, and bot-generated status messages; and records attributed to verified bot accounts. After such filtering, 4,323 records retain (ERC-8004: 142; Google A2A: 4,181). SHA-256 checksums for all raw data fields are provided in the authors' GitHub repository [1]. See also Appendix A.

3.2.2 Annotation. The retained records was annotated by MiniMax-M2.5 [38] with four categorical fields:

- **Stakeholder Institution**: Google / MetaMask / Ethereum Foundation / Coinbase / Independent / Unknown
- **Argument Type**: Technical / Governance-Principle / Economic / Process / Off-topic
- **Stance**: Support / Oppose / Modify / Neutral / Off-topic
- **Consensus Signal**: Adopted / Rejected / Pending / N/A

Institution labels were assigned through a two-tier cascade that prioritizes higher-confidence sources: manual review for the top 109 contributors (yielding 40 verified upgrades), and LLM inference for the remaining 571 authors. The cascade is detailed in Appendix C.

[1] https://github.com/kl41r3/erc8004-a2a-case-study

3.3 Discourse Composition Analysis

To characterize the topical and argumentative composition of the two governance discourses we apply three methods of increasing inductive depth.

3.3.1 Supervised argument typing. Treating the LLM-assigned *argument_type* as the record's communicative function, we test cross-case independence with a chi-square test [12] and within-ERC-8004 temporal change across three consecutive two-month phases spanning submission to mainnet deployment (phase boundaries in Appendix B).

3.3.2 BERTopic comparative discourse analysis. Following Stine and Agarwal [54], we fit BERTopic [22] jointly on the combined corpus. Texts are embedded with `all-MiniLM-L6-v2` [46], reduced via UMAP ($n_{\text{neighbors}} = 15$, cosine, seed= 42) [37], and clustered with HDBSCAN (min cluster size 10) into $K = 19$ topics plus a noise class. For each topic t and case c, the within-case share is

$$p_t^{(c)} = n_t^{(c)} / n^{(c)}.$$

Cross-case divergence is measured by Jensen–Shannon divergence [35]:

$$\mathrm{JSD}(p, q) = \tfrac{1}{2}\mathrm{KL}(p, m) + \tfrac{1}{2}\mathrm{KL}(q, m)$$

with

$$\mathrm{KL}(p, m) = \sum_i p(i) \log \frac{p(i)}{m(i)}, \qquad m = \tfrac{1}{2}(p + q),$$

and JSD = 0 denotes identical distributions, JSD = 1 disjoint.

To assess whether ERC-8004's topic concentration is an artefact of a general-purpose embedding model rather than a genuine structural property, we also embed the 142 ERC-8004 records with CryptoBERT, a model further pre-trained on cryptocurrency social-media posts (StockTwits, Reddit, Telegram, Twitter) [30].

3.3.3 Thematic-LM inductive themes. To complement the embedding based view with human-interpretable labels, we apply the method of Thematic-LM [50], a four-stage multi-agent pipeline: (1) *open coding* assigns a short code to each record; (2) *aggregation* groups 300 sampled codes into 14 raw clusters; (3) *codebook review* merges and refines to 19 themes (T01–T19); (4) *theme assignment* labels every record with its best-fitting theme, or *Unclassified* when confidence is insufficient. We again report per-theme shares and JSD between cases.

3.4 Relational Network Analysis

We construct three complementary networks for each case, each layer adding discursive information to the previous. All graphs use actors as nodes; what changes is the semantic content of an edge.

3.4.1 Co-participation network (SNA). Following Ao et al. [5], an undirected edge connects two contributors who both posted to the same discussion thread (forum topic, GitHub issue, pull request, or discussion); edge multiplicity counts co-occurrences. We report six structural measures:

- Density: the realized fraction of possible ties. Computed by

$$\rho = \frac{2E}{N(N-1)}$$

Table 1: Positioning of this study relative to prior work, organized by paper type. Each panel applies an evaluation rubric appropriate to its class; the synthesis line at the bottom describes how this study bridges all three.

(a) Foundational / Seminal Works — evaluated by methodological or conceptual contribution.

Paper	Year	Core Contribution	Inheritance for this Study
Russell [49] *'Rough Consensus' & the Internet–OSI Standards War*	2006	"Rough consensus" as a standardization mode	Frames ERC-8004's decision rule
Roth & Cointet [48] *Social and Semantic Coevolution in Knowledge Networks*	2010	Coevolution of social and semantic ties	Root of our network–discourse layer
Leifeld [33] *Discourse Network Analysis of Policy Change*	2013	Discourse Network Analysis (DNA)	DNA layer used in §Methods
Beck et al. [6] *Governance in the Blockchain Economy*	2018	IS framework: rights / accountability / incentives	Defines the decentralisation axis
Grootendorst [22] *BERTopic: Neural Topic Modeling*	2022	Neural topic modelling with c-TF-IDF	Used in our topic-discovery pipeline

(b) Perspective / Opinion Papers — evaluated by central claim and the empirical backing offered.

Paper	Year	Central Claim	Domain	Backing
Murray et al. [41] *Contracting in the Smart Era*	2021	Smart contracts and DAOs reshape agency cost	DAO×Corp	Theory
Lumineau et al. [36] *Blockchain Governance: A New Way of Organizing*	2021	Blockchain as a new mode of organizing	DAO×Corp	Theory
Hui & Tucker [25] *Decentralization, Blockchain, AI*	2025	Decentralisation is the right frame for AI governance	AI-Proto	Conceptual
Reineke et al. [47] *Decentralization: A Revolution or a Mirage?*	2025	Decentralisation is contested between revolution and mirage	DAO×Corp	Review
Sunyaev et al. [55] *From Ideology to Design: Purposeful Decentralization*	2026	Decentralisation should be design-driven, not ideological	DAO×Corp	Conceptual

(c) Empirical / Methodological Studies — evaluated by method, setting, openness, and whether the study performs a cross-governance comparison.

Paper	Year	Method	Setting	Data	Code	Cross-gov?
Stine & Agarwal [54] *Comparative Discourse Analysis via Topic Models*	2020	Comparative topic models	Policy discourse	×	×	✓
Ao et al. [5] *Is DeFi Actually Decentralized?*	2023	Social network analysis	Aave (DAO)	×	✓	×
Wu et al. [44] *Decoding Social Sentiment in DAO*	2024	Sentiment + discourse	6 DAO forums	×	×	×
Wang et al. [56] *Understanding DAOs: Governance Dynamics*	2025	Scaled empirical analysis	Snapshot DAOs	✓	×	×
Chen et al. [11] *Governance Tokens & Steemit (PSM-DID)*	2025	Quasi-experimental PSM-DID	Steemit (98 K users)	×	×	×
Shi et al. [50] *Thematic-LM: LLM-Based Thematic Analysis*	2025	LLM-aided thematic analysis	Generic corpora	×	✓	×

This study (2026) *DAO vs. Corporate Governance in AI Protocol Standardization* — inherits the methodological lineage of panel (a); subjects panel (b)'s claims to empirical test on a matched pair; and extends panel (c) with the first cross-governance comparison: 4,323 governance records across ERC-8004 and Google A2A, joint topic discovery (BERTopic + Thematic-LM), three layered network analyses (co-participation SNA, DNA, socio-semantic bipartite), with open data and open code.

where E denotes the number of edges and N denotes the number of nodes.

- Degree Gini:inequality of interaction counts (d_i is node degree). Computed by

$$G = \frac{\sum_{i,j} |d_i - d_j|}{2N \sum_i d_i}$$

where d_i is the number of edges of node i. $G = 0$ implies perfectly equal participation; $G \to 1$ implies that interactions are concentrated in a small elite.

- Components and giant-component ratio: fragmentation into disconnected threads. Computed by

$$\mathrm{GCR} = N_{\max}/N.$$

- Newman–Girvan modularity: reported for institution partitions and Louvain partitions [8] [42]. Computed by
$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$
where m is the total number of edges, A_{ij} is the adjacency matrix entry, k_i is the degree of node i, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same community, 0 otherwise.
- Core–periphery: Borgatti–Everett coreness [9]. Computed by
$$\rho_{\mathrm{BE}} = \mathrm{corr}(A, \Delta), \qquad \Delta_{ij} = \delta_i \cdot \delta_j$$
where $\delta_i \in \{0, 1\}$ is the coreness label of node i. Statistical significance is evaluated by comparing the observed ρ_{BE} against a null distribution of random graphs that preserve the empirical degree sequence (configuration model), following the q-s test of Kojaku and Masuda [29].
- Betweenness centrality and network efficiency: identifies governance brokers—actors who mediate between otherwise disconnected groups:
$$b(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$
where σ_{st} is the total number of shortest s–t paths and $\sigma_{st}(v)$ the number that pass through v.
- Network efficiency: mean normalised harmonic centrality, measured by
$$\bar{h} = \frac{1}{n(n-1)} \sum_{u \neq v} d(u, v)^{-1}$$
where $d(u, v)^{-1} = 0$ for unreachable pairs [31].

3.4.2 Discourse Network Analysis (DNA). Following Leifeld [33], we construct the co-participation layer with stance-aware edges that capture shared discursive positions. Let A be the actor set and $T = \{T_{01}, \ldots, T_{19}\}$ the Thematic-LM codebook. For each record we encode stance into Support= +1, Modify= +0.5, Neutral= 0, Oppose= −1 and obtain an actor–theme stance matrix M similar to the pseudo-matrix below:

	Theme A	Theme B	Theme C
Alice	+1	−1	—
Bob	+1	+0.5	0
Carol	—	−1	+1

Two networks are projected from M: the *congruence network* G^+ connects actors who take same-sign stances on at least one shared theme (Alice and Bob agree on Theme A), and the *conflict network* G^- connects actors who take strictly opposite-sign stances (Alice and Bob conflict on Theme B); edge weights count the number of themes satisfying each criterion. We report density, Louvain modularity and betweenness centrality $b(v)$ of the top-5 discourse brokers.

3.4.3 Socio-semantic bipartite network. Following Roth and Cointet [48], we construct a two-mode (bipartite) network $\mathcal{B} = (A \cup T, E)$, where A is the set of actors, T the set of Thematic-LM themes, and an edge $(a, t) \in E$ exists whenever actor a authored at least one record assigned to theme t. The weight B_{at} equals the number of such records, forming a non-negative integer matrix $B \in \mathbb{Z}_{\geq 0}^{|A| \times |T|}$.

Two one-mode projections are derived from B. The *actor–actor projection* $W^A = BB^\top$ links any two actors by the number of themes they co-discussed (self-loops removed); the *theme–theme projection* $W^T = B^\top B$ links any two themes by the number of actors who engaged in both.

Per-actor topic diversity is measured by Shannon entropy:
$$H(a) = -\sum_{t=1}^{|T|} \hat{p}_{at} \log_2 \hat{p}_{at}, \qquad \hat{p}_{at} = \frac{B_{at}}{\sum_{t'} B_{at'}},$$
where $H(a) = 0$ for a pure specialist (all posts in one theme) and $H(a) = \log_2 |T|$ for a perfect generalist (uniform spread across all $|T|$ themes).

We further characterise the distribution of $\{H(a)\}_{a \in A}$ via its Gini coefficient. The higher the Gini, the more thematic breadth is concentrated in a few actors. We also measure per-theme actor concentration as the Gini of the column vector $\{B_{at}\}_{a \in A}$ for each t.

The thematic overlap coefficient is measured by
$$\Omega = \frac{|T_1 \cap T_2|}{\min(|T_1|, |T_2|)},$$
where T_c denotes the active theme set of case $c \in \{1, 2\}$, quantifies cross-case thematic alignment; $\Omega = 1$ means one community's thematic space is a subset of the other's.

Actor-set sizes across the three filtering stages are detailed in Appendix D.

4 Results

4.1 Decision Architectures

ERC-8004 and Google A2A represent contrasting governance archetypes.

On the left of figure 1 showd the ERC-8004 governance process [1]. There are 3 types of EIPs: standard track EIP, Meta EIP and Informational EIP. The standards track EIP can be broken down to 4 categories: core, networking, interface and ERC [16]. All EIPs, including ERC-8004 in our case, undergo 5 stages: idea, draft, review, last call and final. EIPs updated continuously are assigned to the special stage of "living". All EIPs are decided by rough consensus: Core EIPs by Ethereum core developers at the meeting called *AllCoreDevs*, and others by common developers and users. ERC-8004 remains at the stage of "draft" when the data was fetched (Mar 2026), nevertheless, its three canonical registry contracts were deployed on Ethereum mainnet on January 29, 2026 [18].

This reflects a structual feature of ERCs as application-layer specifications: unlike Core EIPs, which modify the Ethereum protocol itself and requires hard-fork activation when deployment, ERCs define smart-contract interface standards that any party may implement independently, regardless of the proposal's formal lifecycle status [16].

On the right of figure 1 reveals the A2A governance structure. The protocol launched under `google/A2A` in April 2025 and was donated to the Linux Foundation in June 2025 [21], migrating to the vendor-neutral `a2aproject/A2A` organization. Governance authority rests with an eight-seat Technical Steering Committee (TSC), comprising representatives each from Google, Microsoft, Cisco, AWS, Salesforce, SAP, IBM, and ServiceNow. Independent contributors cannot join the TSC during an 18-month startup phase [3].

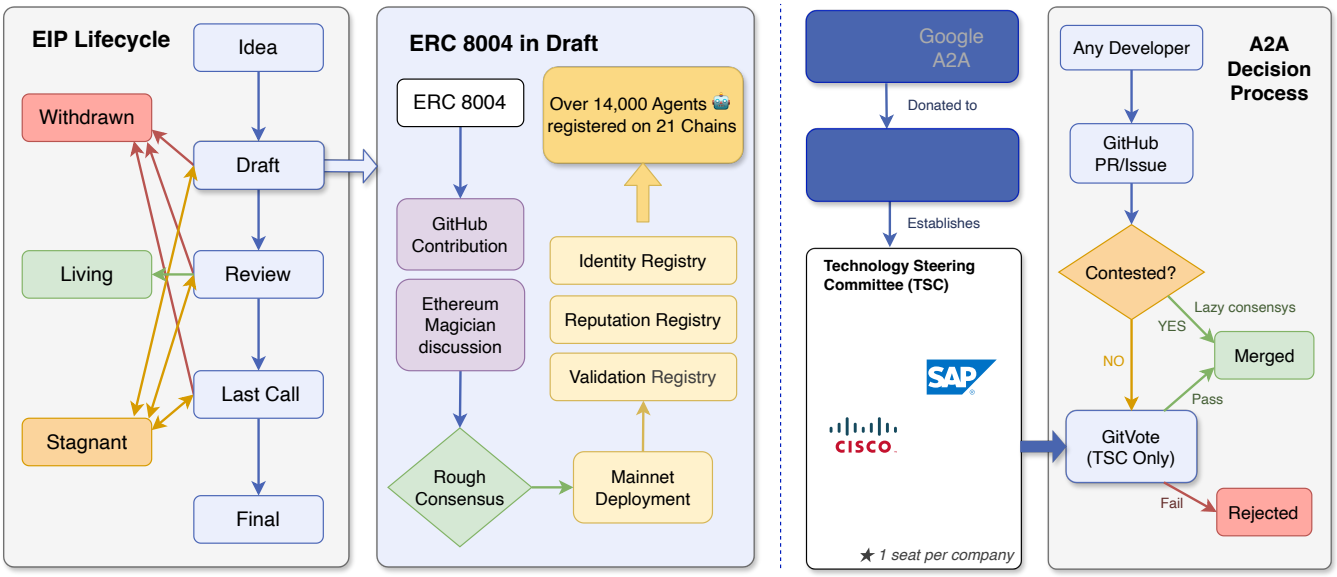

Figure 1: Left: ERC-8004 governance decision flow. Proposals are discussed through the Ethereum Magicians forum and are recorded on GitHub. Right: A2A governance structure and decision flow. The protocol transitioned from Google-controlled to Linux Foundation governance. Contested decisions result in a GitVote.

Routine pull requests are merged upon maintainer approval without a formal vote [3]; contested specification changes escalate to a GitVote [4], a GitHub-native ballot in which only TSC members cast binding votes at a 51% threshold.

Algorithm 1 EIP Lifecycle

Data: $Stages = \{IDEA, DRAFT, REVIEW, LAST_CALL, FINAL, STAGNANT, WITHDRAWN\}$
$stage \leftarrow IDEA$
while $stage \neq FINAL$ **and** $stage \neq WITHDRAWN$ **do**
  Proceed to $NEXT(stage)$ based on community consensus;
  if No activity for a long period **then**
    $stage \leftarrow STAGNANT$;
  else if Author decides to cancel **then**
    $stage \leftarrow WITHDRAWN$;
  end if
end while
Deployment: Theoreticaly any party *MAY* deploy registry independently of *stage*

Algorithm 2 A2A Decision Process (TSC Governance)

Input: GitHub Pull Request or Issue (pr)
Output: Decision Status (Merged or Rejected)
if pr is NOT Contested **then**
  // Apply Lazy Consensus
  $Status \leftarrow MERGED$;
else
  // Escalation to Technology Steering Committee (TSC)
  Initiate **GitVote** (Restricted to 1 seat per company);
  if Vote == PASS **then**
    $Status \leftarrow MERGED$;
  else
    $Status \leftarrow REJECTED$;
  end if
end if
return $Status$;

4.2 Discourse Composition

4.2.1 Supervised argument typing. Figure 2 compares the LLM-assigned argument type distributions. Both communities are predominantly Technical (74.3% in ERC-8004 vs. 62.1% in A2A), but A2A devotes almost twice the share to Process arguments (25.4% vs. 13.9%). The cross-case difference is significant with a small effect size ($\chi^2(3) = 52.88$, $p < .001$, Cramér's $V = .103$): both cases remain technically grounded, but corporate governance carries visibly heavier coordination overhead.

Within ERC-8004 (Figure 3), argument composition shifts significantly across three two-month phases ($\chi^2(6) = 25.32$, $p < .001$, $V = .315$). Technical arguments dominate Phases 1–2 (≥80%); in Phase 3 Process discussion surges to 53% as deliberation moves from substantive design to editorial ratification. This two-stage pattern—technical convergence followed by procedural ratification—is consistent with rough-consensus norms in open standards bodies. Small-n caution applies to Phases 2–3 ($n = 11$ and $n = 17$).

Figure 4 cross-tabulates stance with argument type. Support and Modify stances are dominated by Technical arguments in both cases (ERC Support 73%; A2A Support 87%): substantive engagement is predominantly technical regardless of governance form. Governance-Principle arguments, particularly under *Oppose*, are

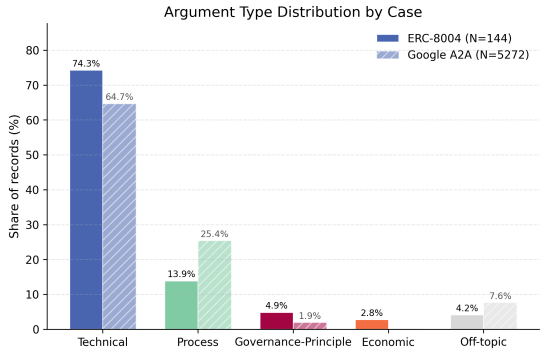

Figure 2: Cross-case argument-type distributions.

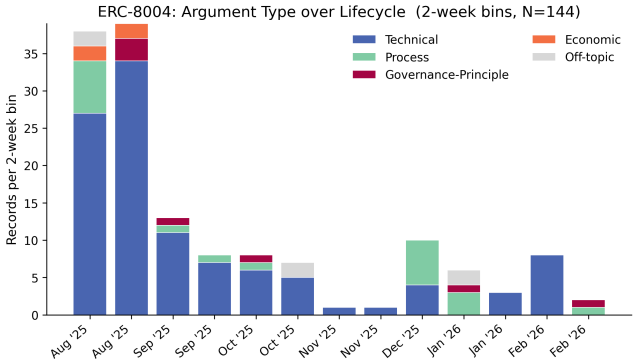

Figure 3: ERC-8004 argument-type distribution across three lifecycle phases.

more visible in ERC-8004, reflecting the principled deliberation characteristic of the EIP process. The sharpest cross-case contrast appears in the *Neutral* row: 44% of A2A Neutral records are Process in nature, against 26% in ERC-8004—corporate governance generates procedural commentary even among non-committal participants.

4.2.2 BERTopic. BERTopic identifies 19 topics plus a noise class over the combined corpus. The global Jensen–Shannon divergence between the two cases' topic distributions is $\mathrm{JSD}_{\mathrm{BERTopic}} = 0.288$, a moderate but meaningful structural separation (Figure 5). ERC-8004 is strikingly concentrated: 67.6% of its records fall into Topic 0 (*agent, agents, for, of*), the broad agent-discourse cluster, versus 21.2% for A2A. Implementation-heavy topics—Task/Message management (8.2%), JSON/proto spec (7.7%), PR-contribution workflows (7.2%), SDK samples (5.9%)—carry substantial A2A weight but zero ERC-8004 records, confirming that low-level engineering is absent from the EIP forum. The only topic with higher ERC share than A2A is Topic 9 (*suggestion, sap, linkedin*; 2.8% vs. 0.9%), signalling corporate voices (SAP, LinkedIn) entering the EIP thread.

CryptoBERT on ERC data shows more precise result. It divides the "agent" topic to T0 (Onchain, Reputation) and T2 (Trust, Feedback), and the total percentage is 73.2%, which aligns with the 67.6% of pure BERTopic.

4.2.3 Thematic-LM. The Thematic-LM pipeline yields a 19-theme codebook (T01–T19). Figure 7 maps the full codebook across cases (left: record share heatmap; right: actor-participation butterfly) and Table 2 reports the six themes with the largest cross-case divergence. The global Jensen–Shannon divergence is $\mathrm{JSD}_{\mathrm{Thematic\text{-}LM}} = 0.216$.

The most discriminating theme is **T08 Trust & Security Mechanisms**, which accounts for 34.5% of ERC-8004 records but only 4.0% of A2A records ($\Delta = +30.5$pp). This theme covers agent-trust scoring, reputation schemes, on-chain credential verification, and vouch-chain models—the foundational problem of establishing trustworthy agency in a permissionless environment. By contrast, A2A spreads deliberation across **T06 Documentation & Examples** (10.8%), **T18 Clarifications & Information Requests** (10.2%), **T07 Community Collaboration & Contributions** (9.9%), and **T01 Protocol Specification & Versioning** (7.9%). T09 (Transport & Protocol Mechanisms) and T16 (Streaming & Real-time Communication) are entirely absent from ERC-8004.

Table 2: Thematic-LM themes with largest cross-case divergence.

ID	Theme	ERC %	A2A %	Δ
T08	Trust & Security	**34.5**	4.0	+30.5
T01	Protocol Specification	**13.4**	7.9	+5.5
T06	Documentation & Examples	2.1	**10.8**	−8.7
T05	SDK Development	0.7	**4.6**	−3.9
T15	Tooling & Automation	0.7	**4.5**	−3.8
T09	Transport Mechanisms	0.0	**3.2**	−3.2

The two methods agree on the direction and approximate magnitude of divergence, with BERTopic's JSD = 0.288 slightly exceeding Thematic-LM's JSD = 0.216. The gap is expected: Thematic-LM compresses divergence by absorbing related concerns into shared conceptual themes, while BERTopic retains finer-grained lexical distinctions (ERC's "trustless", "reputation" versus A2A's "samples", "proto"). Together the two estimates bracket the structural divergence and confirm its robustness: the two governance processes occupy meaningfully different but overlapping discourse spaces, with ERC-8004 concentrated on constitutive layer of the design space ("what to build and why") and A2A on executive ones (how to build, document, and ship it).

4.3 Relational Networks

4.3.1 Co-participation network (SNA): core and perifery. Table 3 reports structural metrics for the co-participation networks (ERC-8004: $N = 67$; Google A2A: $N = 771$). Figure 8 visualizes both networks; the A2A panel shows the top-50 contributors for readability, while all metrics use the full population.

Both communities exhibit high degree inequality. The Gini coefficient of degree is 0.804 for ERC-8004 and 0.779 for A2A, indicating that in both cases, a small elite drives the majority of interactions. The top-3 contributors account for 32.3% of all ERC-8004 interactions and 14.9% of A2A interactions. In A2A, two out of three of Google employees, and another is from Microsoft. In ERC-8004, the three contributors are: MarcoMetaMask (Marco De Rossi), who obviously by the name is from *Metamask*; spengrah (Spencer Graham), cofounder of *Hats Protocal*; and pcarranzav (Pablo Carranza

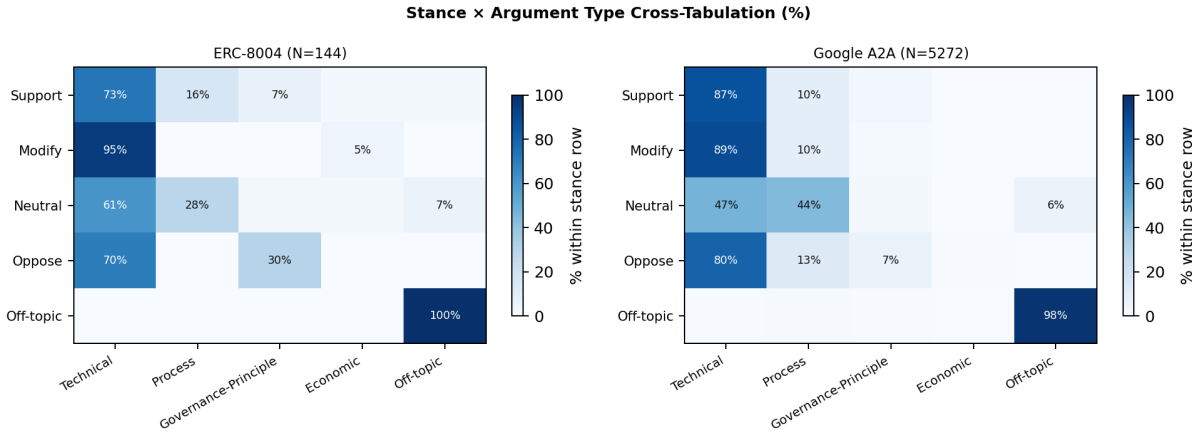

Figure 4: Stance × argument-type cross-tabulation (% within stance row). ERC-8004 (N = 144, left) and Google A2A (N = 5,272, right).

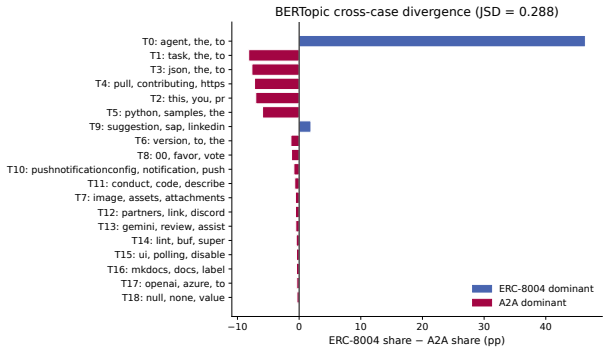

Figure 5: BERTopic cross-case divergence. Topics are sorted by absolute difference; blue bars indicate ERC-8004 dominance, orange bars indicate A2A dominance. Global JSD = 0.288**.**

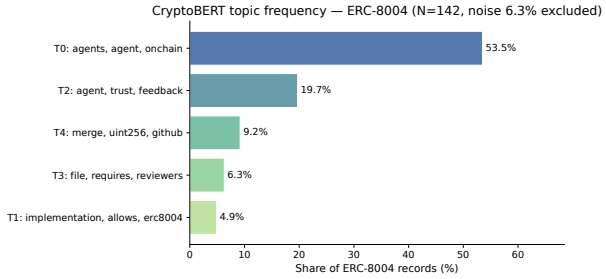

Figure 6: CryptoBERT topic frequency for ERC-8004 records (N = 142, noise class 6.3% excluded). Bars shaded light-to-dark by topic rank; the two dominant topics (T0 and T2) together account for 73.2% of records, consistent with BERTopic's broad agent-discourse cluster.

Table 3: Governance Network Structural Metrics

Metric	ERC-8004	Google A2A
Nodes	67	771
Edges	65	1,230
Density	0.029	0.004
Gini (degree)	0.804	0.779
Top-3 degree share	32.3%	14.9%
Components	43	346
Giant component ratio	0.328	0.534
Modularity—institution	−0.059	−0.034
Modularity—Louvain	0.425	0.473
Louvain communities	46	358
CP p-value (BE test)	0.095	1.000
CP significant ($p < .05$)	No	No
Top-1 betweenness centrality	0.069	0.136
Betweenness Gini	0.931	0.979
Top-3 betweenness share	70.8%	48.5%
Network efficiency $\bar{h}$	0.050	0.110

Vélez), who is from *The Graph*, a decentralized protocal for indexing and querying blockchain data. It is not determined whether they represent their organizations or they partitipate as independent contributors. A possible answer is that, they participate in both identities. On the one hand, they are driven by strategic objectives; one the other hand, they earn themselves social reputation in the open-source world.

Both networks are structurally fragmented and none shows statiscally significant core-perifery structure. Louvain community detection recovers 46 communities for ERC-8004 and 358 for A2A, closely matching the component counts and confirming that participation organises around parallel threads rather than a coherent deliberative body. Institution labels also do not predict interaction structure in either case. The giant component covers 32.8% of ERC-8004 nodes and 53.4% of A2A nodes; 333 of 771 A2A participants

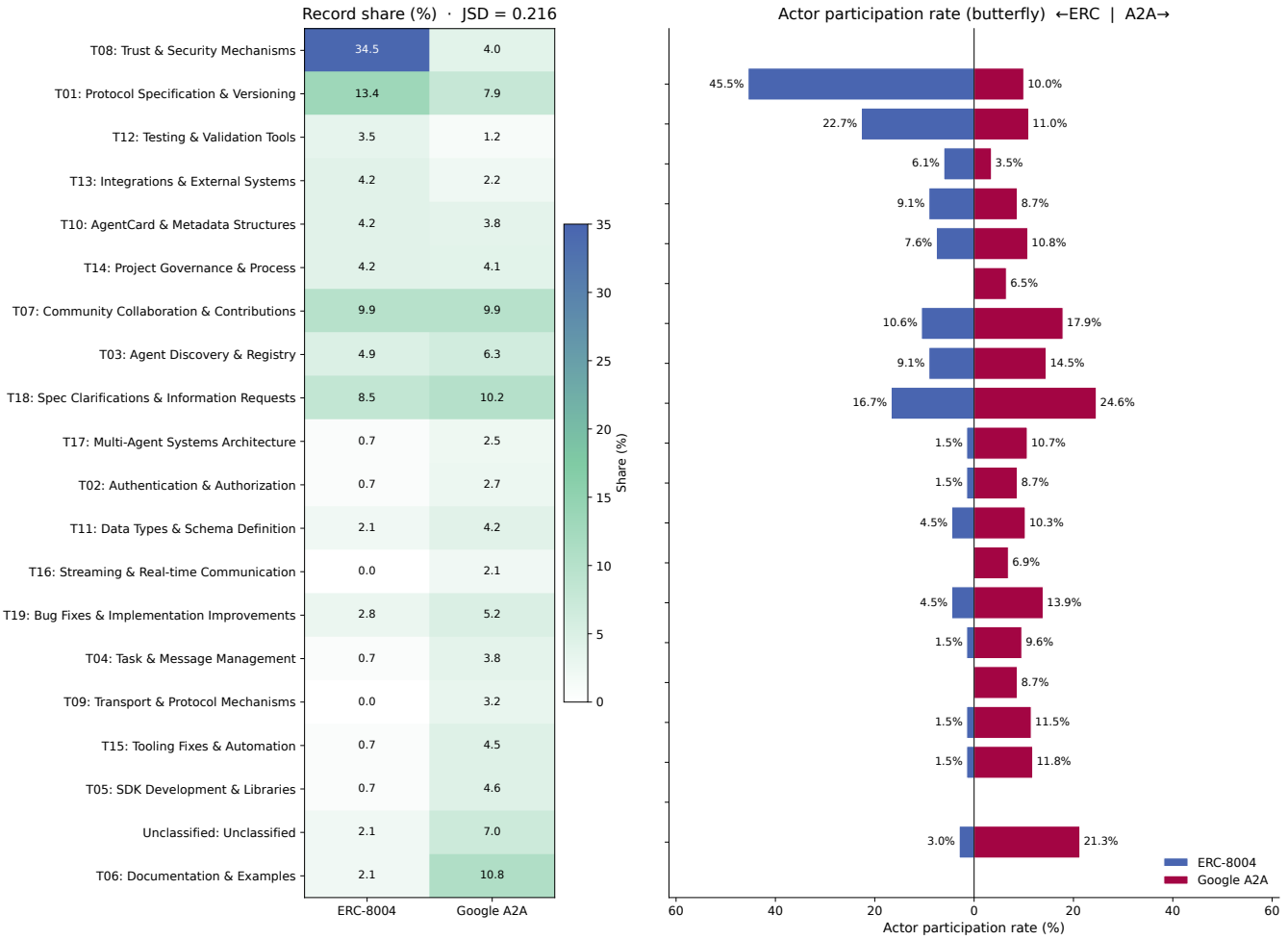

Figure 7: Left: Thematic-LM per-case record share (%), rows sorted by $\Delta = p^{\mathrm{ERC}} - p^{\mathrm{A2A}}$ (ERC-dominant themes at top). Right: Per-theme actor-participation rate (butterfly view); for each theme the left bar is the share of ERC-8004 actors who engaged it at least once, the right bar is the equivalent for A2A. Blue = ERC-8004; red = Google A2A.

(43.2%) are complete isolates with no co-participation edges. Density indicates that the vast majority of possible co-participation pairs are unrealised. Under the Borgatti-Everett test [9], ERC-8004 returns $p = .095$ and A2A returns $p = 1.000$; neither crosses the $\alpha = .05$ threshold.

4.3.2 Discourse network: congruence and conflict. The stance layer (Table 4) reveals structures that are invisible in the co-participation graph.

ERC-8004 achieves denser within-community consensus while A2A generates far greater absolute conflict volume. Congruence density is 0.148 for ERC-8004 versus 0.082 for A2A: within the tighter EIP community, participants more often share positions on the same theme. Conflict edges in A2A are 34× larger in absolute count (2,531 vs. 74). This disparity stems not only from the scale effect of 710 versus 66 actors, but also from the broader range of technical positions supported by a multi-vendor engineering project.

Table 4: Discourse Network Analysis metrics.

Metric	ERC-8004	Google A2A
Actors	66	710
Active themes	17 / 19	19 / 19
Congruence edges	318	20,638
Congruence density	0.148	0.082
Congruence modularity (Louvain)	0.2886	0.2453
Top-1 betweenness centrality	0.061	0.018
Betweenness Gini	0.859	0.889
Top-3 betweenness share	34.5%	12.2%
Conflict edges	74	2,531
Mean actor theme diversity	1.470	2.211

The top-3 betweenness share in the congruence network drops to 34.5% for ERC-8004 and 12.2% for A2A, compared with 70.8% and 48.5% in the co-participation network (Table 3). Discourse brokerage is more distributed than structural interaction brokerage in

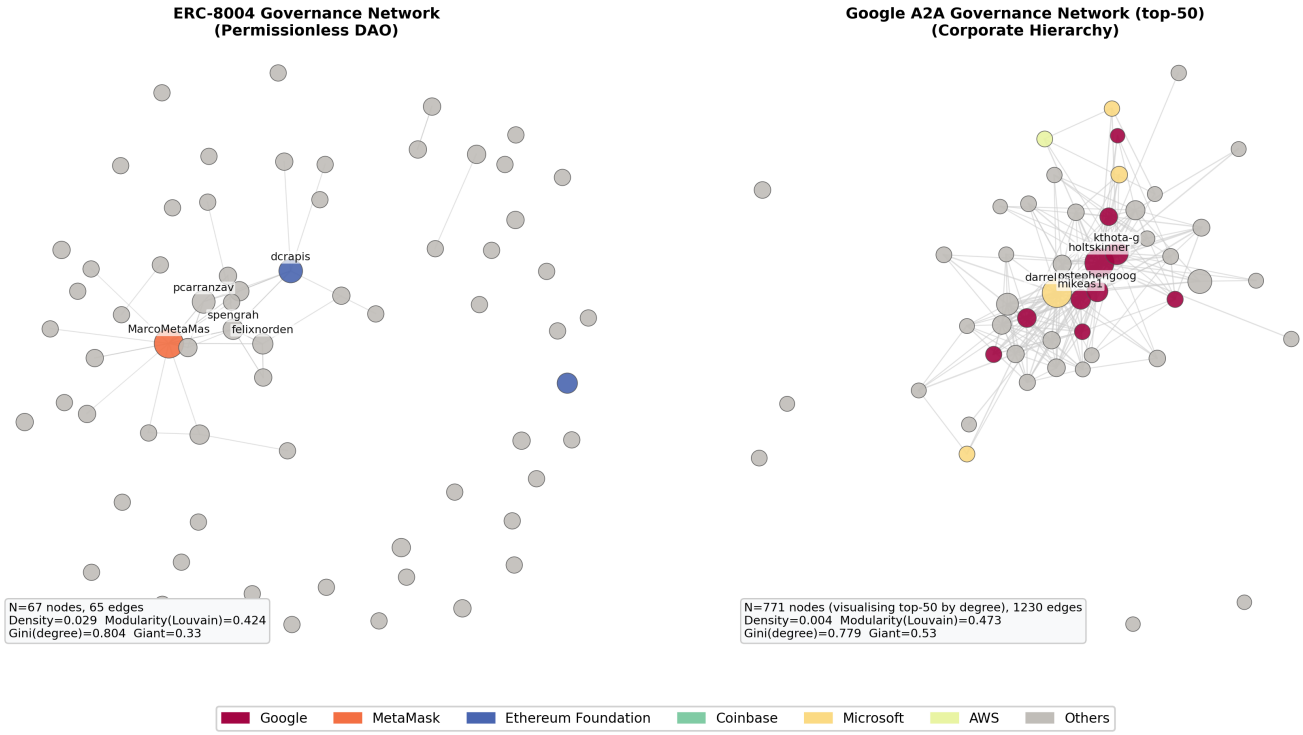

Figure 8: Governance interaction networks. ERC-8004 (left, $N = 67$, all contributors) and Google A2A (right, top-50 contributors shown for readability; all structural metrics computed on $N = 771$). Node color encodes institutional affiliation; node size encodes contribution count.

both cases; yet ERC-8004's congruence network remains markedly more concentrated, consistent with a compact EIP core mediating contested positions on trust and protocol security.

4.3.3 Socio-semantic bipartite network: thematic concentration of discursive labor. The actor–theme bipartite layer (Table 5, Figure 9) reveals where each governance community directs its deliberative effort. Peripheral participation is the structural baseline in both cases: median actor Shannon entropy $H = 0$ in both ERC-8004 and A2A, meaning the majority of contributors engage a single theme across their entire record [39]. This is not a governance-specific feature but an intrinsic property of large-scale deliberative systems.

Table 5: Socio-semantic bipartite network metrics.

Metric	ERC-8004	Google A2A
Actors	66	710
Active themes	17/19	19/19
Actor–actor projection edges	645	59,007
Mean actor entropy H	0.348	0.617
Median actor entropy H	0	0
Gini(H)	0.773	0.707
Max H	2.664	3.834
Mean theme Gini (actor concentration)	0.085	0.453
Most-discussed theme	T08	T06
Thematic overlap Ω	1.000	

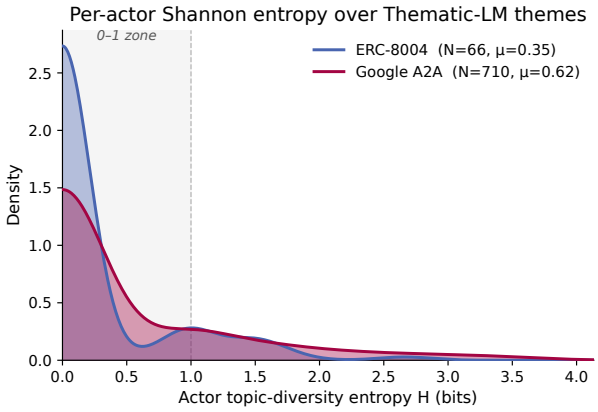

Figure 9: Per-actor Shannon entropy over Thematic-LM themes. $H = 0$ marks pure specialists; the $H = 0$ bin dominates both distributions.

What differs across governance forms is *which* themes absorb deliberative effort, and how broadly the most active participants range. Mean actor entropy is 0.348 for ERC-8004 and 0.617 for A2A: A2A's core contributors span roughly twice as many themes on average. Gini(H) remains high in both cases (0.773 vs. 0.707),

confirming that thematic breadth is itself concentrated in a small number of generalist actors in both communities.

The cross-case divergence is sharpest at the theme level. In ERC-8004, 34.5% of actors participated in T08 (Trust & Security Mechanisms), and 13.1 particiated in T01 (Protocol Specification & Versioning). For A2A, the pattern inverts for engineering-execution themes: Documentation & Examples (T06) engaged 10.8% of A2A actors but only 2.1% of ERC-8004 actors. Mean theme actor-concentration Gini is 0.453 for A2A against 0.085 for ERC-8004, indicating that each A2A theme attracts a narrower, more dedicated subgroup.

The thematic overlap coefficient $\Omega = 1.0$: all 16 active ERC-8004 themes reappear in A2A. A2A additionally covers three themes without ERC-8004 actor participation: T09 (Transport & Protocol Mechanisms), T14 (Project Governance & Process), and T16 (Streaming & Real-time Communication). T09 and T16 are purely engineering-execution concerns entirely absent from the EIP forum; T14's ERC-8004 records were authored by automated review bots and thus fall outside the actor network. This confirms that DAO governance discourse concentrates on constitutive design questions (security mechanisms, protocol principles), while corporate governance additionally commands the execution layer.

5 Discussion and Future Directions

The three sub-questions return a coherent picture. The two governance regimes differ sharply in formal decision architecture (RQ1) and in *what* their communities deliberate about (RQ2), yet converge on *how* their communities are relationally organized—both exhibit comparable participation inequality, fragmented community structure, and absence of statistically significant core–periphery (RQ3). We draw three theoretical implications for CSCW research on the governance of collaborative infrastructure.

5.1 Decentralization as Design, Not as Fact

Decentralized governance is appealing for its idealistic design of distributed authority and trustless coordination. In practice, however, the ERC-8004 process exhibits a centralization paradox composed of two distinct mechanisms.

First, routine decisions are not subjected to voting, one of the most governence processes in DAOs. The power to decide, for example, which proposals are "ready", which objections are "substantive", which positions count as "rough consensus", accumulates in the hands of a few minorities. In our data this manifests as the top three contributors accounting for 32.3% of all interactions despite no formal authority delegation, mirroring the participation inequality patterns repeatedly documented in voluntary online cooperation [13, 20, 39, 43].

Second, no formal hierarchy is provided to distribute power of decision. One possible reason for this is that the small scale of community size don't request such hierarchy, however one direct drawback is people tend to groupthinking, resulting in the denser discourse congruence. The Ethereum Magicians forum has no Wikipedia-style closing protocol for resolving stalled disputes [27], no constitutional mechanism comparable to A2A's GitVote escalation [3], and no documented norms governing who is entitled to declare consensus.

These two features compound: in the absence of formal procedure, situation falls to the most natual case: the certain minority holds the voice. The decentralization promised by the EIP architecture exists at the level of entry rights; in practice, routine decision authority concentrates around whoever has the time and reputation to remain in the room.

By contrast, A2A's TSC encodes oligarchy explicitly: eight named seats, one vote per company, transparent rules of escalation [3]. The A2A discourse network distributes betweenness more evenly (top-3 share 12.2% in congruence, against 34.5% for ERC-8004) precisely because external contributors must engage the formal pull-request workflow rather than wait out a vote.

5.2 The Confounding Role of Open Source

A natural objection is that the convergence we observe stems not from governance form but from open-source norms shared by both cases. ERC-8004 and A2A are both public and permissive-licensed projects. The 1% rule of online participation [43], the heavy-tailed contribution distribution of OSS [7, 13, 19, 39], and the structural realities of contemporary open-source ecosystems [20, 57] all predict the participation pattern we report regardless of whether the formal governance label is "DAO" or "corporate TSC".

We treat this not as a refutation but as a delimitation. First, the topical divergence between the two cases (JSD = 0.288 on BERTopic, 0.216 on Thematic-LM) cannot be reduced to OSS norms; it is governance-driven, reflecting the constitutive–regulative distinction between proposing a standard (what the protocol *is*) and shipping an implementation (*how* it is built). Second, ERC-8004's discourse-congruence density is nearly twice that of A2A, the opposite of what an OSS-scale-only account would predict.

The cautious framing that follows is therefore layered: open-source publication establishes a baseline of high participation inequality and fragmented community structure; governance form then redirects which themes this skewed participation engages and which actors accrue informal authority. CSCW work that combines institutional and socio-technical analysis [57] provides the right register for this layered reading, and our results extend that line into the AI-protocol domain by holding the technical task constant across two governance regimes.

5.3 Limitations and Future Directions

Three limitations bound our claims. First, ERC-8004 contributes 142 records (3.3% of the retained corpus) against A2A's 4,181, so themes of low frequency carry wide confidence intervals and small-n caution applies to ERC-8004's phase-level results ($n = 11$ and $n = 17$ in Phases 2 and 3). Second, ERC-8004 deliberation is by design recorded on public forums and GitHub, whereas A2A's TSC meetings, internal Google design reviews, and partner negotiations occur outside the public repository. The structural concentration we observe in A2A may therefore understate deliberation among its core members. Third, the open-source confound discussed above: both cases are public voluntary projects, and the structural similarities we measure are partly attributable to open-source norms.

These limitations point to productive directions. Comparing closed-source corporate consortia against public DAOs would isolate the open-source effect from the governance effect, despite that

data accessibility may be a problem. Tracking governance participation across a protocol's full lifecycle, including post-deployment maintenance, would reveal whether the Phase-3 attrition we observe in ERC-8004 reverses or deepens after mainnet. Whether the governance revolution that decentralization proponents promise can be fulfilled through better incentive design [11], purposeful governance architecture [55], or institutionalised democratic legitimacy [40] remains an open and consequential question for AI agent protocols now entering mainnet.

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A Data Provenance

All raw data files are versioned with SHA-256 checksums stored in `data/raw/CHECKSUMS.json` in the project repository. Table 6 lists the file names, record counts, and collection dates for the primary raw data files used in this study.

Table 6: Raw Data Files and Record Counts

File	Records	Collected
forum_posts.json	113	2026-03
github_comments_filtered.json	36	2026-03
a2a_issues.json	3,104	2026-03
a2a_prs.json	1,955	2026-03
a2a_discussions.json	822	2026-03
Total (raw)	6,030	
Total (retained)	4,323	

B ERC-8004 Lifecycle Phase Boundaries

Topic analysis within ERC-8004 was conducted across three consecutive two-month phases spanning the full proposal lifecycle (2025-08-13 to 2026-01-29):

- **Phase 1** (Aug 13 – Oct 13, 2025): Initial submission and early community review. $n = 100$ records.
- **Phase 2** (Oct 13 – Dec 13, 2025): Sustained review period prior to Last Call designation. $n = 11$ records.
- **Phase 3** (Dec 13, 2025 – Feb 13, 2026): Last Call through Final ratification (mainnet deployment 2026-01-29). $n = 17$ records.

C Institution Label Provenance Cascade

Institution labels were assigned using the following three-tier cascade, applied in priority order:

(1) **Manual investigation** (high confidence): A systematic review of GitHub profiles, LinkedIn pages, personal websites, and EIP commit metadata was conducted for the top 109 contributors across both cases (all 71 ERC-8004 participants plus the top 38 A2A contributors by post count). This review yielded 40 institutional upgrades, replacing LLM-inferred labels with verified affiliations. The original LLM-inferred label is preserved in a separate `institution_lm` field for all records to enable sensitivity checks.
(2) **LLM inference** (lowest confidence): For the remaining 517 authors, institution was inferred by MiniMax-M2.5 from contextual signals in the record text (e.g., repository ownership, self-identification, email patterns where visible).

D Actor Filtering Stages

The three network analyses operate on successively filtered actor sets (Table 7). Starting from the annotated records (ERC: 142 records / 71 contributors; A2A: 4,181 / 778), the co-participation network drops four ERC and seven A2A bot/short-text residuals, yielding $N = 67$ and $N = 771$ respectively. The DNA and socio-semantic analyses further (a) inner-join against the Thematic-LM `coded_records.json` (12 records lose a theme assignment) and (b) exclude records whose stance is *Off-topic* or *Unclassified*, dropping actors whose *only* contributions fall into those categories. This produces the final stance-bearing actor sets of $N = 66$ (ERC-8004) and $N = 710$ (Google A2A). All SNA statistics use $N = 67$ / $N = 771$; all DNA and socio-semantic statistics use $N = 66$ / $N = 710$.

Table 7: Actor and record counts through the three filtering stages.

	ERC-8004		Google A2A	
	Records	Actors	Records	Actors
Annotated (retained)	142	71	4,181	778
SNA (bot & short-text filter)	130	67	4,230	771
DNA / Socio-semantic	126	66	3,759	710